

Health Shocks, Social Insurance, and Firms*

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Abstract

We study the role that firms play in social insurance benefit uptake after their workers experience health shocks. Social insurance in our setting, Hungary, is universal and comprehensive, thus allowing us to quantify the impact of firms on benefit uptake and labor market outcomes on top of the social safety net. Using linked employer-employee administrative data with individual-level health records, we find that firm responses to worker health shocks are heterogeneous: workers hit by a health shock at high-quality firms are less likely to take up disability insurance or exit the labor force than those at low-quality firms.

Keywords: health shock; disability insurance; firm heterogeneity

JEL Codes: H55, I10, J22, J23

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1 Introduction

Do firms shape how workers fare after a major health shock, on top of what a comprehensive social insurance system already provides? We study this question in Hungary, where health insurance coverage is close to universal and disability insurance (DI) is available to any worker who suffers a health shock, regardless of employer. Because the safety net is uniformly available to everyone, any systematic differences across firms in workers' employment and DI outcomes after a health shock cannot be explained by differences in the generosity or availability of benefits: they must instead reflect something about the firm itself.

We answer this question using linked employer-employee administrative data from Hungary covering 2003–2017, matched to individual-level health records available from 2009. We identify major health shocks as a worker's first hospitalization of at least ten days following at least two years without hospitalization, and trace employment and DI status for three years before and after the shock. Firm quality, following Abowd, Kramarz, and Margolis (1999), is measured as a firm's wage premium net of observable worker and job characteristics; we ask whether a worker's post-shock trajectory depends on which tertile of this firm-quality distribution they were in when the shock hit.

We find that health shocks have a larger impact on workers in lower-quality firms. Three years after a health shock, the impact of health shocks on DI takeup is 64 percent larger for workers in the bottom tertile of the firm quality distribution than for those in the top tertile, and the impact on loss of employment is 32 percent larger on the bottom than on top. Furthermore, workers in higher-quality firms face a higher probability of falling down the ladder than those in lower-quality firms: while they are 24 percent more likely to remain employed, they are indeed employed in firms of the bottom two tertiles of the quality distribution. We show evidence that these patterns are not driven by the frequency or type of health shocks experienced by workers of different firms, by differences in pre-shock health level (proxied by healthcare use), by the sorting of workers across firms along the quality distribution, or by the lower replacement rate of disability benefits for high-wage workers.

Our results suggest that firms play an important role in mediating the consequences of health shocks for their workers. Conditional on experiencing a health shock, a worker's likelihood of leaving employment and taking up DI depends significantly on where they work. Job security after a health shock is, in this sense, a workplace amenity like any other. A large literature has documented that "good" firms—those that pay wage premia—often also offer better working conditions along other dimensions: safety (Lavetti, 2020), flexibility (Mas and Pallais, 2017), paid time off (Maestas, Mullen, Powell, von Wachter, and Wenger, 2023), family leave (Bana, Bedard, Rossin-Slater, and Stearns, 2023), and the option to work from home (Barrero, Bloom, Davis, Meyer, and Mihaylov, 2022), consistent with a broader view of compensating differentials across jobs (Rosen, 1986; Sorkin, 2018). Because higher-wage jobs tend to come bundled with better amenities (Hotz, Johansson, and Karimi, 2018; Sockin, 2022; Bana, Bedard, Rossin-Slater, and Stearns, 2023; Bíró and Elek, 2025; Lachowska, Mas, Saggio, and Woodbury, 2026), measures of inequality that account for amenities are typically larger than measures based on wages alone.

Beyond paying higher wages, then, high-quality firms also appear to offer better protection against the consequences of health shocks, which implies that earnings inequality understates overall inequality in worker welfare. The direct policy implication is that firm-targeted policies such as experience rating could reduce DI uptake by addressing this heterogeneity, though their incidence would fall disproportionately on lower-quality firms, whose workers take up benefits at higher rates.

Our work contributes to multiple strands of the literature. We most directly contribute to the literature that has considered the role of firms in disability insurance (DI) takeup. This literature so far is inconclusive: while Kyrrä and Paukkeri (2018) show no major effect of experience rating on the inflow to sickness and disability benefits, Hawkins and Simola (2024) and Prinz and Ravesteijn (2020) provide evidence for significant impacts of experience rating on disability inflow. Our results suggest that after a major health shock, even in the absence of experience rating, the inflow to DI varies substantially with firm quality. This means that experience rating the DI system may address heterogeneous responses to health shocks and thus incentivize the retention of workers following these shocks. This implies that firm-side policies—part of many policy proposals (Autor and Duggan, 2010; Autor, 2015; Burkhauser and Daly, 2011; Liebman, 2015)—could play an important role in disability insurance.

More broadly, we contribute to a recent strand of the literature that has examined the relationship between firm characteristics and the takeup of various social insurance programs. Considering temporary benefits, the use of which is generally considered a positive outcome, Bana, Bedard, Rossin-Slater, and Stearns (2023) show that temporary DI and paid family leave program takeup is substantially higher at firms with high earnings premiums. Aizawa, Mommaerts, and Rennane (2022) show that firm characteristics explain a substantial part of variation in the takeup of disability accommodations. Lachowska, Sorkin, and Woodbury (2025) focus on unemployment insurance (UI) claims and provide evidence for important employer effects. They argue that some of the incomplete UI claiming is due to experience rating of the UI payroll tax. We contribute to this literature by focusing on labor market and social security outcomes after a major health shock, and analyzing the interplay of individual and firm quality in these outcomes. We perform this analysis in a setting where social insurance is universal and comprehensive.

Our paper is also related to the literature on the impact of health shocks on individuals' labor market outcomes (e.g., Dobkin, Finkelstein, Kluender, and Notowidigdo, 2018; Fadlon and Nielsen, 2021; Parro and Pohl, 2021).¹ This literature generally estimates negative effects of health shocks on employment and earnings, although the estimates vary. We are not familiar with studies that provide evidence on heterogeneities in the labor market consequences of major health shocks by firm quality.²

¹For an overview of this literature, see Section 3.4 of Prinz, Chernew, Cutler, and Frakt (2018).

²In recent work, Ahammer, Packham, and Smith (2023) show that firms are responsible for nearly 30 percent of the variation in across-worker healthcare expenditures but they do not analyze the consequences of health shocks across firm types. In a related study, Bíró and Elek (2025) find that moving to higher-quality, higher-paying firms increases employees' use of preventive and diagnostic healthcare—particularly for cardiovascular conditions—without worsening physical health, while reducing mental health medication use for men and older

2 Background, data, and methods

Hungary is a useful setting for isolating the role of firms in benefit uptake precisely because its social insurance system is close to universal. Health insurance coverage of the population is close to 100 percent (Gaál, Szigeti, Csere, Gaskins, and Panteli, 2011) under a unified public scheme funded from payroll taxes, and public inpatient and outpatient care are free of charge to insured individuals; medications are free for hospital patients, though pharmacy purchases require a co-payment. Employees who are not able to work due to health reasons are entitled to 15 days of sick leave per year, during which they receive 70 percent of their salary from their employer. Afterwards, they are entitled to sickness benefit (50–60 percent of their salary) for up to one year, paid by social security. While workers are on sick leave or sickness benefit, the employer is not allowed to lay them off. People with health damage of at least 40 percent can apply for disability insurance (DI), the amount of which depends on the severity of the health problem and previous earnings. During the period we study, individuals were allowed to work while receiving DI benefits, subject to an earnings limit of 150 percent of the minimum wage. Recipients of the rehabilitation allowance for beneficiaries with less health damage were allowed to work 20 hours per week. Benefits are financed from social security contributions that are not experience-rated, i.e., they do not depend on the share of employees of a firm claiming benefits. People with at least 20 years of work history are entitled to old-age pension benefits upon reaching the statutory retirement age. Over our analyzed period, the statutory retirement age increased from 62 to 63.5 both for men and women. However, women could retire earlier with 40 years of work credit, regardless of age.

We combine this institutional setting with linked employer-employee administrative data covering years 2003–2017 on a random 50 percent sample of the 2003 population. Our sample is drawn from the whole population, and not just those who have a job. Health records in the data cover the years 2009–2017, and DI and old-age pension status are observed in 2003–2016. The database consists of linked data sets at the monthly frequency of the pension, tax and health care authorities and contains detailed individual-level information on employment and earnings history, use of the health care system, pension and other social benefits, and firm-level indicators. Importantly, it also contains information on the type and amount of different disability benefits and old-age pensions received.³ An individual is defined to be employed in a given month if he is employed on the 15th of the month, and the employment is not under the public works scheme. We also observe in the data the employer of the worker (including self-employment). For employers with double-entry bookkeeping we also observe balance sheet information from the tax authority. We restrict the sample to ages 18–60, so that workers close to or above the statutory retirement age are excluded.

In these data, we define health shocks as the first month of hospitalization for a previously unhospitalized individual. Specifically, we generate the binary indicator of a health shock by

workers, consistent with better workplace health amenities and less stressful working conditions.

³Two important limitations of the data are that the employment status of DI recipients cannot be observed until April 2007 and we do not observe the health condition based on which the disability benefit is received.

first flagging 12-month periods with at least 10 days of hospital stay. In such 12-month periods, we identify the first month of hospitalization. We set the health shock indicator to one for the first month of hospitalization, conditional on no hospitalization during the preceding 24 months. We set the health shock indicator to zero if there was no hospitalization either during the current and next 11 months or the preceding 24 months. We apply the 10-day cut-off in the length of hospital stay to ensure that we focus on major health problems that potentially lead to the DI uptake.

In the following, we work with annual data, retaining, for each year, the observation corresponding to the calendar month of the first hospitalization during the shock period.⁴ For each individual we consider only the first health shock event. For individuals without any health shocks, the observation month is generated randomly. Since health-related data is available only from 2009 in the administrative data we use, the health shock indicator is defined only for years 2011–2017 to have at least 2 years of pre-shock observations. For women, we define health shocks the same way as for men, but do not consider birth-related hospitalizations as health shocks. Note that while we use the term “health shock” to indicate hospitalization without prior hospitalization, these events may be expected (or planned) based on existing health problems. Nevertheless, our health shock indicator captures a major health event (prolonged hospital stay).

To measure firm quality, we perform an Abowd, Kramarz, and Margolis (1999, AKM)-style wage decomposition and compute worker and firm wage premia (fixed effects, FE). That is, we regress wages on individual and firm fixed effects, controlling for year fixed effects, age squared, age cubed,⁵ (firm-specific) tenure, tenure squared and firm size. We also include a large set of occupation dummies to account for wage heterogeneity by occupations, similarly as in Boza (2021). For individuals who experience a health shock, we split their wage observations into a healthy and an unhealthy set (before and after the shock) in order to avoid the negative effect of health shocks on wages entering the worker effects. We estimate the following wage regression:

$$\ln w_{ijt} = \mathbf{X}_{ijt}\beta + \theta_{ih(it)} + \psi_{j(it)} + \lambda_{k(it)} + \delta_t + \varepsilon_{ijt}, \quad (1)$$

where i is an individual, j is a firm, t indicates time, k represents occupations, and h refers to the health status of an individual (healthy or not). In our main analysis we use the person effect related to the healthy period of all individuals.⁶ When estimating the worker and firm wage premia, we include all sample years of the linked employer-employee administrative data and also include both men and women in the sample. Employment always includes self-employment but excludes public-sector work, and it is weighted by working hours (e.g., if someone works 20 hours per week then that is counted as 0.5 employment). For most of our analyses, we divide firms into three tertiles (low, middle, and high FE firms).⁷ These firm fixed effects capture more

⁴We replicate our main analysis on monthly data and show similar patterns.

⁵We follow Card, Cardoso, Heining, and Kline (2018) and assume that the wage profile is flat at the age of 41.

⁶The worker premium estimated for the unhealthy period of a given individual is on average 78.5 percent of that of the healthy period of the same individual, with a strong correlation of 0.81.

⁷Limited mobility bias, a common issue in the AKM framework, is not a concern in our setting. This bias would

than pure firm-specific productivity. The literature has identified several additional sources of firm wage premia, including labor market power (Berger, Herkenhoff, and Mongey, 2022; Lamadon, Mogstad, and Setzler, 2022), product market power (De Loecker, Eeckhout, and Unger, 2020; Edmond, Midrigan, and Xu, 2023), and non-wage amenities (Wong, 2026).

Our monthly wage measure includes all the income that forms the basis of social security contributions. We focus on the wages of full-time workers (working weekly 40 hours, which holds for 87 percent of the working population in our analysis sample), and deflate the wages to 2015 using the CPI. To ensure that the wage patterns are not driven by lower wage during sickness absence, we adjust the wage for sickness benefit receipt. We also adjust the wage for sick leave (first 15 days of sickness absence in a given calendar year), assuming that the 15 days of sick leave are used in the month preceding the first month of sickness absence, or in January, if the worker already has sickness absence in January. Because of likely measurement errors in the adjusted monthly wage, we use the average of non-zero wages in a given calendar year as our preferred wage measure.

Before turning to our main results, we present evidence that workers do not select into firms based on their health status. The probability of a health shock is similar across calendar years and months, with the exception of a higher probability in January and an offsetting lower probability in December, consistent with some hospitalizations being reallocated from the end of one year to the start of the next (Appendix Figure A1). The distribution of hospital days and days of work absence following a shock is similar across all three firm-quality categories; the majority of workers who experience a shock spend fewer than 20 days in hospital and are absent for less than two months over the following year (Appendix Figure A2). The probability of a health shock ranges from 1.0 to 1.4 percent across firm-quality categories, and this difference falls below 0.04 percentage points once we net out age, occupation, and calendar year (Appendix Figure A3). Table 1 shows that workers who experience a health shock are on average 5.7 years older and earn about 8 percent less than those who do not, but that firm quality and the occupation and industry composition of the two groups are similar. Appendix Table A1 shows that the prevalence of major disease categories at the onset of a shock, led by cardiovascular, digestive, musculoskeletal, and urogenital conditions and accidents, is essentially identical across firm-quality tertiles. Figure 1 previews our main result: following a health shock, DI entry rises and employment falls by the most at the lowest-quality firms. Some sorting of workers across firms by quality is present, as shown in Appendix Table A2, and is similar for workers with and without a health shock; more than 10 percent of the sample nonetheless falls into the low-firm/high-individual or high-firm/low-individual quality cells, providing meaningful variation for our analysis.

cause some firms to be incorrectly classified into to wrong tertile—if anything, this would shrink the difference in firm group-level outcomes. Even so, we exclude low-mobility firms—those with less than two mobility events per year, either inflow or outflow—to mitigate this issue.

[Table 1 around here]

[Figure 1 around here]

To quantify the impact of the health shock on labor market outcomes, we estimate the following regression:

$$Y_{it} = \sum_{\substack{j=-3 \\ j \neq -1}}^3 \alpha_j \mathbb{1}[E_{it} = j] + \sum_{\substack{j=-3 \\ j \neq -1}}^3 \beta_j \mathbb{1}[E_{it} = j] \cdot S_i + \xi age_{it}^2 + \mu_i + \delta_t + u_{it}, \quad (2)$$

where i indexes individuals, t indexes calendar year, E_{it} indicates event time (in years), which is the time to the health shock or to the random event for the control group (individuals never experiencing a health shock). Y_{it} is a labor market outcome indicator, μ_i captures individual fixed effects⁸, δ_t captures calendar year effects, and S_i is a binary indicator for ever experiencing a health shock. Event time for people who do not experience a health shock is defined by assigning a random health shock year and month. We restrict the estimation sample to individuals employed full time (40 hours per week) in the private sector at event time zero. The coefficients of interest are the β 's which capture the differential evolution of labor market outcomes for individuals who suffer a health shock (treatment group) relative to individuals who do not experience a health shock (control group).

To analyze the heterogeneity in outcomes by firm quality, we allow the β parameters to vary with the tertiles of AKM firm FE at event time zero (we denote this indicator with F_i ranging from 1 to 3) and control for AKM firm FE tertile at event time zero:

$$Y_{it} = \sum_{\substack{j=-3 \\ j \neq -1}}^3 \alpha_j \mathbb{1}[E_{it} = j] + \sum_{l=1}^3 \sum_{\substack{j=-3 \\ j \neq -1}}^3 \beta_j^l \mathbb{1}[F_i = l] \cdot \mathbb{1}[E_{it} = j] \cdot S_i + \xi age_{it}^2 + \mu_i + \delta_t + u_{it}. \quad (3)$$

3 Results

We now turn to discussing our results. Workers impacted by health shocks are, unsurprisingly, more likely to take up DI and less likely to remain employed than those without shocks. More striking, however, is a steep gradient by firm quality. As Table 2 shows, the impact on DI takeup is 64 percent larger for workers at low-quality firm than at high-quality ones (12.5 vs. 7.6 percentage points 3 years after the shock), while the employment impact is roughly a quarter, 24 percent smaller for workers at high-quality firms (8.1 vs. 10.7 percentage points).⁹

[Table 2 around here]

⁸Our results are robust to omitting the individual fixed effects but adding age, sex, and the AKM person effect ($\theta_{ih(it)}$ from equation (1)) as control variables.

⁹Appendix Figure A4 plots the estimated regression coefficients over time, and Appendix Figure A5 displays them on a monthly frequency. Appendix Figure A6 shows similar patterns by firm quality quintiles, instead of tertiles.

Decomposing the employment effect by destination reveals an intriguing pattern. Among workers at high-quality firms, the smaller overall employment decline masks two opposing forces: a negative impact on continued employment at the same firm, and a positive impact on employment at a different firm. Specifically, shocked workers at high-quality firms are 2.6 percentage points more likely to be employed at a different firm relative to the unshocked, whereas the corresponding effect is negative for workers at lower-quality firms. Workers receiving health shocks at high-quality firms find new employment predominantly at lower-quality firms: the combined effect on their probability of working at a lower-quality firm is 3.1 percentage points, while the probability of moving up the quality ladder is only 1.1 percentage points for those at low-quality firms. The likelihood of moving down the firm quality ladder is thus nearly three times as large as that of climbing it.

Taken together, these results point to workers at high-quality firms being better insulated against health shocks, as they can remain employed after a health shock even if it requires moving to a lower-quality firm. Workers already at the bottom of the quality distribution, however, face both a steeper employment penalty and a substantially higher rate of DI takeup.

A natural concern is that these patterns reflect treatment effect heterogeneity across shocked workers rather than the firm-quality gradient. Appendix Figure A7 dispels this concern, showing that our results are robust to implementing the estimator of Sun and Abraham (2021) which allows for the presence of treatment effect heterogeneity across treatment cohorts. Note that we report these estimates only for the DI outcomes because we cannot apply the estimator for the employment outcomes. The reason is that the Sun and Abraham (2021) estimator uses the control group to identify the time trend without attaching an event time; however, our control group for the employment outcomes must be employed at event time zero which would be undefined here. The heterogeneity we document is similarly present when we restrict the sample to full-time workers three years before the shock and measure firm quality three years before the shock as well, rather than at the shock itself (Appendix Figure A8).

Because different types of firms may employ different types of workers, one might worry that our results simply reflect how different kinds of workers respond to a health shock, rather than a causal role for the firm. We cannot rule out this channel entirely, as sorting of better workers to better firms is a basic feature of the labor market; however, Appendix Figures A9 and A10 show that the same pattern holds even within a given tertile of individual AKM fixed effect, and within a given broad occupation category: holding worker productivity fixed, the effect of a health shock is still smaller at higher-quality firms. Nor can the pattern be explained by the fact that DI has a lower replacement rate for high earners, who tend to work at higher-quality firms. The replacement rate falls from 78 percent in the bottom decile of relative wage to 25 percent in the top decile, though it is close to flat through the seventh decile (Appendix Figure A11). Re-estimating our baseline regressions on the sample of workers in the bottom half of the relative-wage distribution, where the replacement rate varies little, we continue to find that a health shock is less likely to trigger DI entry and has a weaker negative employment effect for workers at top-tertile firms (Appendix Figure A12); the confidence intervals are wider on this

restricted sample, but the pattern is unchanged.

[Figure 2 around here]

The results are also not driven by differences in the health shocks themselves across firm types. Estimating the average effect of a shock one to three years out separately by major disease category, we find that cancer and cardiovascular disease have the largest positive effects on DI uptake and the largest negative employment effects (Figure 2; Appendix Table A3 reports all outcomes), but the firm-quality gradient holds within essentially every disease category. Healthcare use follows a similar pattern regardless of firm quality: sickness benefit days, GP and specialist visits, and prescription drug spending all rise sharply at the onset of a shock and remain elevated three years later, with little variation across firm-quality tertiles in either pre-shock levels or the size of the post-shock increase (Appendix Figure A13). The gradient is also robust to how we measure firm quality in the first place: using foreign ownership (above 50 percent) or total factor productivity¹⁰ in place of the AKM fixed effect, we find that two to three years after a shock, DI probability is about two percentage points lower at foreign-owned or high-TFP firms than at their counterparts (Appendix Figure A14).

The firm-quality gradient in DI uptake holds across a range of subgroups. It is similar for women and men, though the average effect of a shock on DI uptake is larger for men (Appendix Figure A15). Workers aged 40–60 are roughly three times as likely to enter DI following a shock as workers aged 18–39, and the firm-quality gradient is present within both age groups (Appendix Figure A16). The effect of a shock on DI uptake is more severe outside Budapest, where labor markets are thinner, and this geographic gap is concentrated among workers at the bottom two firm-quality tertiles (Appendix Figure A17). Splitting the sample by the largest industry groups, manufacturing, trade, services, and a residual category, the effect of a shock on DI probability is at least two percentage points lower at top-tertile firms than at the bottom two tertiles in every industry (Appendix Figure A18). Splitting by firm size shows a similar gap, around five percentage points, within every size category, which rules out the possibility that our results are driven simply by larger firms' greater capacity to reassign a shocked worker's tasks (Appendix Figure A19).

A health shock also affects wages, though in the opposite direction across the firm-quality distribution. Figure 3 plots log wages over time by firm quality (Panel a) and the corresponding estimated effects of Equation (3) (Panel b): relative to workers who did not experience a shock, wages fall further behind at higher-quality firms, with an estimated effect of about -1 percent at the bottom tertile compared with -3 to -5 percent at the top two tertiles two to three years after the shock. Wages at higher- and lower-quality firms were already trending in opposite directions before the shock relative to the control group, however, so we are cautious about attributing this pattern directly to the shock itself. The same caveat applies to income, defined as wages plus DI benefit and not restricted to full-time workers, which falls by about 3 percent

¹⁰We calculate the value added-based TFP. When doing so, we apply the estimation procedure of Wooldridge (2009) and use the `prodest` Stata package by Rovigatti and Mollisi (2020).

at top-quality firms and rises by about 1 percent at bottom-quality firms three years after the shock (Appendix Figure A20).

[Figure 3 around here]

Finally, a health shock also affects job mobility and hours, conditional on remaining employed. Workers at lower-quality firms are generally more likely to switch employers than workers at higher-quality firms, but we find no clear heterogeneity in how a health shock affects this baseline propensity (Appendix Figure A21, Panel a). A health shock raises the probability of moving to a new employer by 1–2 percentage points three years out, somewhat more so for workers who started at top-tertile firms, though the estimates are noisy and some pre-trends are present (Panel b); this provides weak evidence that a worker who remains employed at a low-quality firm after a shock is also more likely to remain with the same employer. Weekly working hours fall by more at the bottom-tertile firms, 0.8 hours per week three years after the shock, than at the top two tertiles, 0.4–0.6 hours, again conditional on employment (Appendix Figure A22).

4 Conclusion

This paper studies the labor market and benefit uptake consequences of major health shocks and investigates how these outcomes vary with the quality of the employer at the time of the shock. We show that workers hit by a health shock at a high-quality firm are more likely to remain employed and less likely to take up disability insurance benefits. We provide evidence that these heterogeneity patterns are not primarily driven by the sorting of high-quality workers to high-quality firms, by differences in the type of health shocks by firm quality, or by the lower replacement rate of disability benefits for high-wage workers.

These results suggest that firms play an important role in mediating the consequences of health shocks for their workers. This in turn implies that social insurance policies should take into account firm heterogeneity. For example, an effective approach may be disability insurance experience rating policies that collect higher premiums from firms with more workers who take up benefits, though their incidence would initially fall on lower-quality firms. Furthermore, the findings in this paper imply that inequality in worker welfare is higher than wage inequality due to the unequal distribution of the amenity of protection from the consequences of health shocks. Workers in higher-quality firms make more and are also more likely to remain employed if they are hit by a shock, exacerbating inequality.

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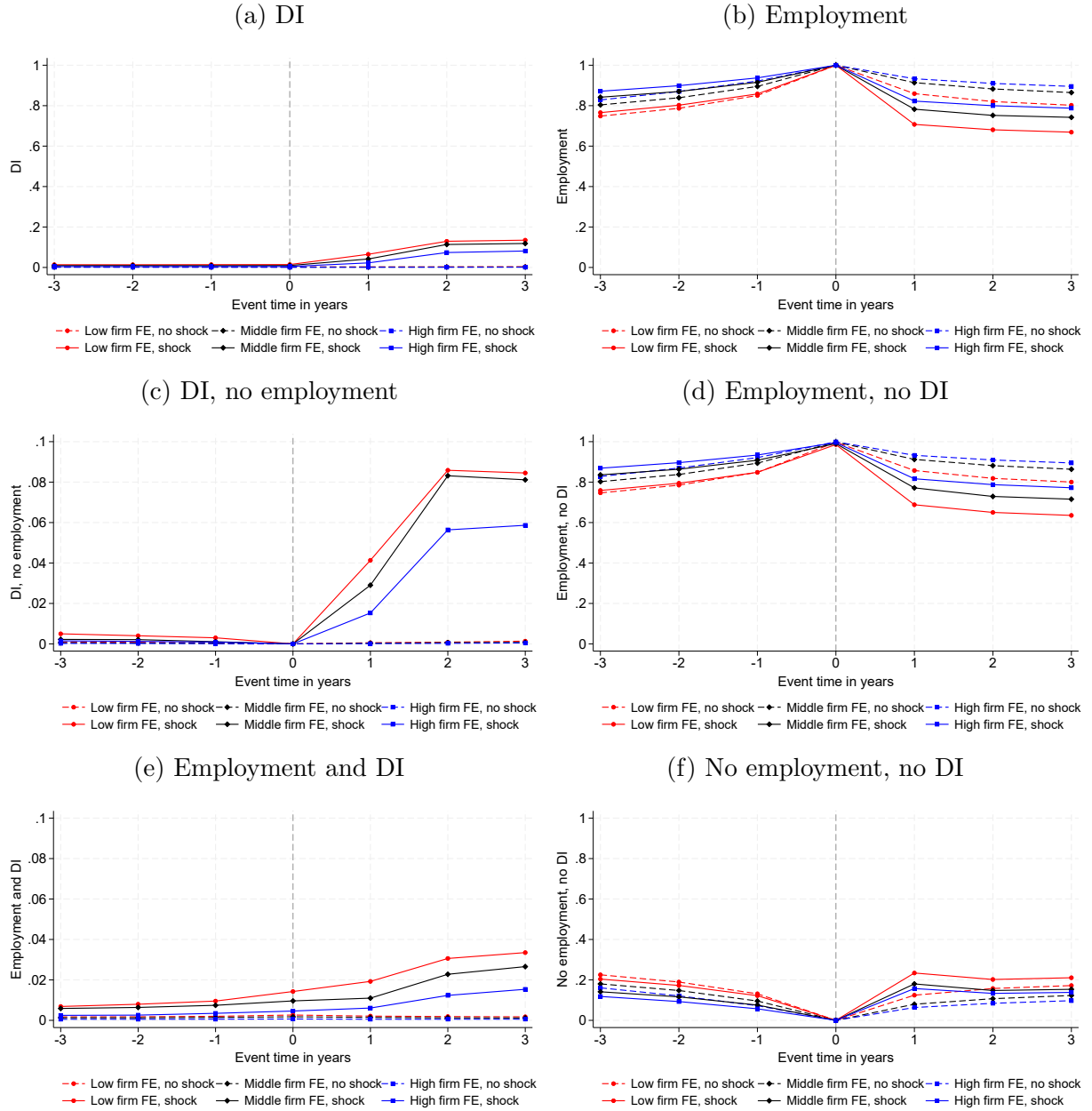
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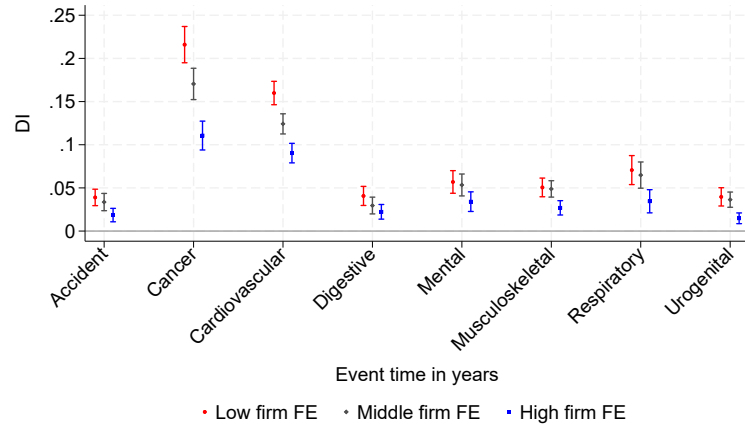
Figures and tables

Figure 1: Labor market outcomes over time by health shock and firm quality



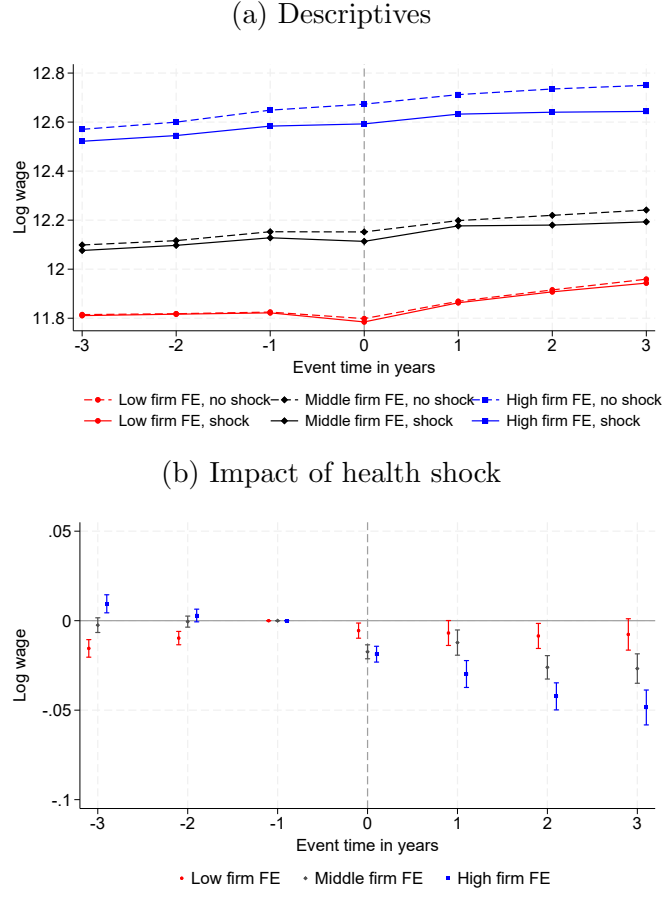
Note: Figure shows time patterns of labor market outcomes for individuals suffering a health shock (treated group) and individuals who never suffered a health shock (control group). For the control group, event time zero is set randomly. Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Figure 2: Average effect of health shocks on DI by disease categories



Note: Figure shows estimated β parameters with 95% confidence interval (based on robust standard errors) of a modified version of Equation (3), in which the event time categories are replaced by a binary indicator that equals zero 1-3 years before the shock and equals one 1-3 years after the shock (the year of the event is omitted). Outcome variable is disability insurance (both with and without employment). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Figure 3: Wages over time by health shock and firm quality



Note: Panel (a) shows time patterns of wages for individuals suffering a health shock (treated group) and individuals who never suffered a health shock (control group). For the control group, event time zero is set randomly. Panel (b) shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed full-time. Firm quality categories refer to the employer at event time zero. Wage is the annual average monthly wage, adjusted for sickness absence. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Table 1: Descriptive statistics

	Health Shock				No Shock			
	Mean	Std.Dev.	p(10)	p(90)	Mean	Std.Dev.	p(10)	p(90)
Men	0.57				0.61			
Age	45.2	9.8	31.0	57.0	39.5	10.1	26.0	54.0
AKM firm FE	0.04	0.23	-0.27	0.35	0.05	0.25	-0.27	0.37
Monthly wage (HUF)	226,610	308,762	107,219	415,570	245,460	410,320	109,549	469,130
Lives in Budapest	0.14				0.16			
<i>Occupation</i>								
Manager	0.07				0.07			
Professional	0.08				0.11			
Other white collar	0.21				0.23			
Skilled blue collar	0.29				0.27			
Assembler, machine op.	0.21				0.19			
Unskilled laborer	0.14				0.12			
<i>Industry</i>								
Manufacturing	0.32				0.32			
Construction	0.05				0.05			
Trade	0.13				0.15			
Accommodation, food	0.03				0.03			
Transportation, storage	0.12				0.11			
Services	0.21				0.22			
Other or missing	0.15				0.12			

Note: Table shows descriptive statistics 12 months before the health shock (first four columns) or 12 months before the health shock indicator is set to zero (i.e., for individuals with zero days of hospitalization in the current and preceding two years – second four columns). The sample is restricted to people employed in the private sector, for whom the occupation code is not missing. The number of observations is 52,010 in the health shock, and 568,478 in the no shock sample.

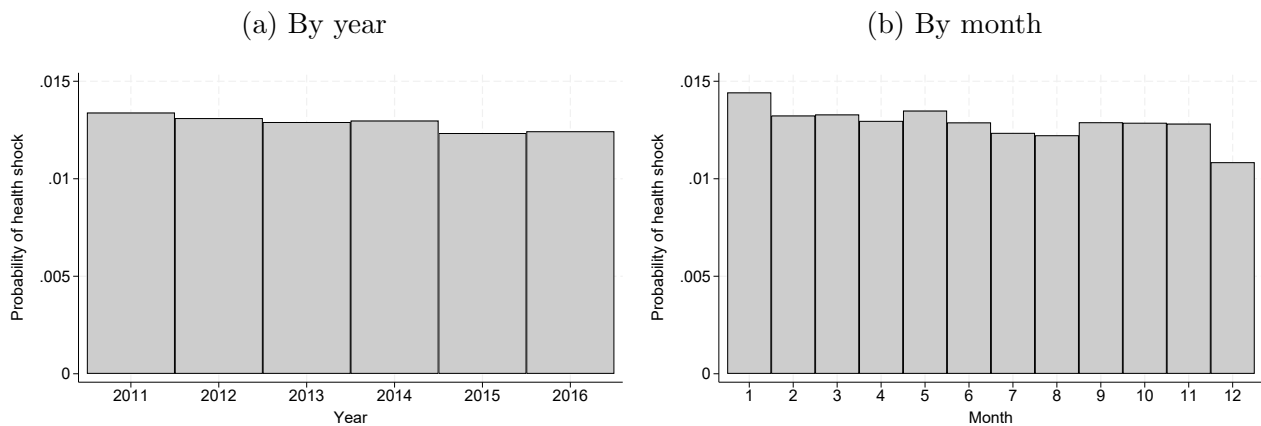
Table 2: Impact of health shocks on labor market outcomes

		DI	Employed	Employed in ...		Employed in different ...		
				same firm	different firm	low-q. firm	middle-q. firm	high-q. firm
Origin firm	Low	0.125***	-0.107***	-0.063***	-0.044***	-0.109***	0.000	0.011***
		(0.004)	(0.005)	(0.006)	(0.006)	(0.006)	(0.003)	(0.002)
	Middle	0.113***	-0.107***	-0.098***	-0.008	0.000	-0.117***	0.002
		(0.003)	(0.005)	(0.006)	(0.005)	(0.004)	(0.006)	(0.003)
	High	0.076***	-0.081***	-0.107***	0.026***	0.009***	0.022***	-0.121***
		(0.003)	(0.005)	(0.006)	(0.005)	(0.002)	(0.003)	(0.006)

Note: Table shows estimated β parameters of Equation (3) three years after the shock and the associated heteroskedasticity-robust standard errors. Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles. Self-employment outcome is not reported.

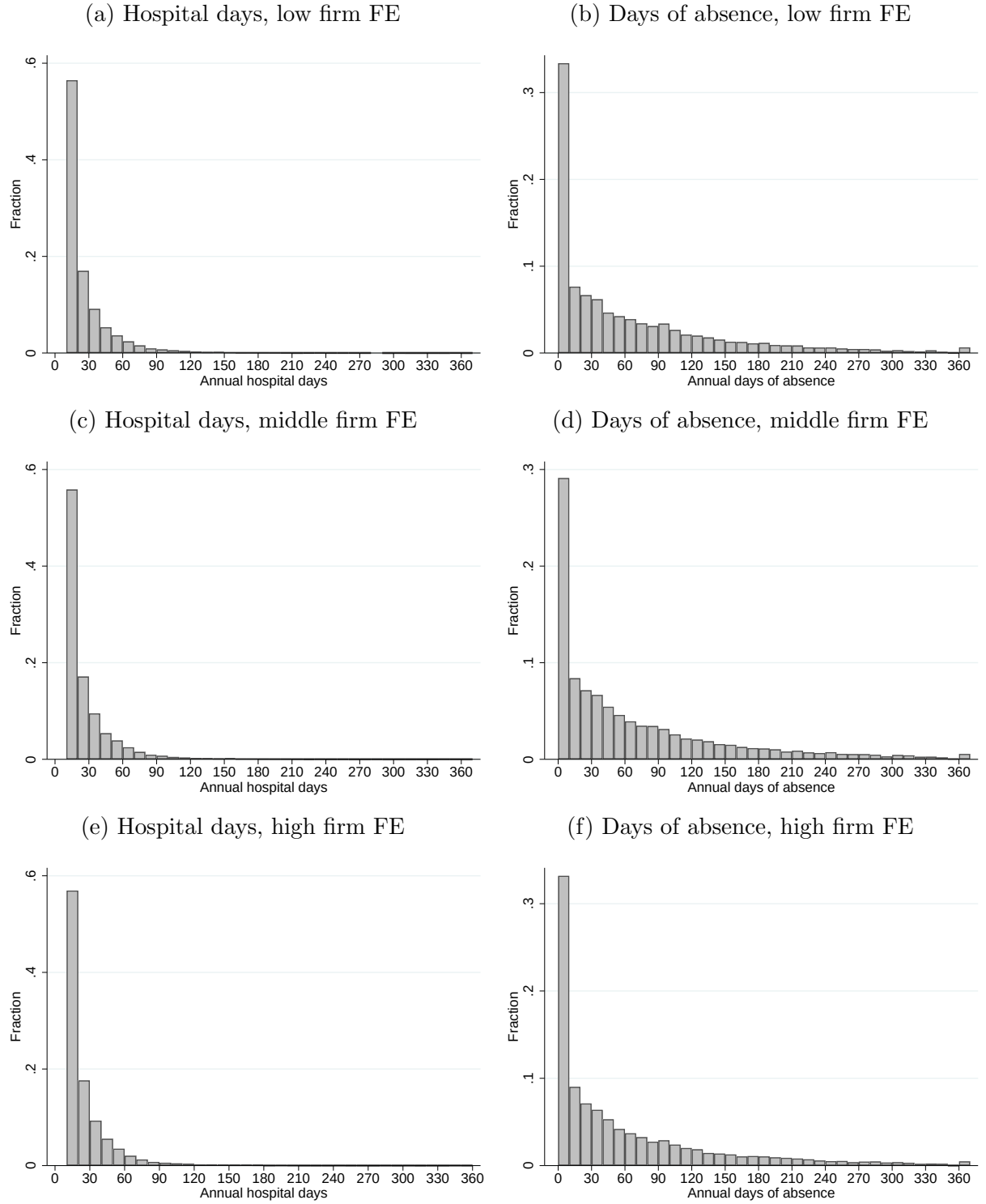
Appendix: Additional figures and tables

Appendix Figure A1: Monthly probability of a health shock by year and month



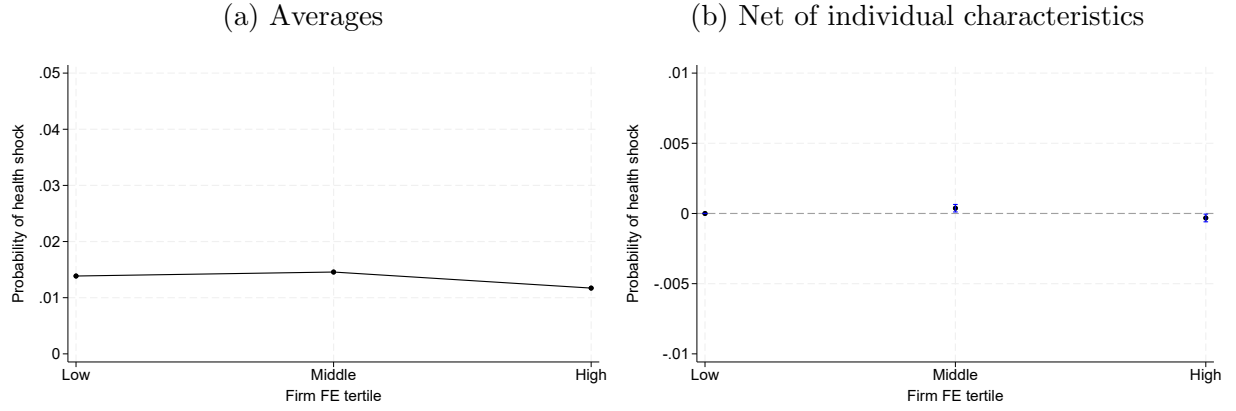
Note: Figure shows the probability of suffering a health shock by calendar year and month over 2011-2016. The health shock definition is provided in Section 2. Sample is restricted to employed people.

Appendix Figure A2: Hospital days and days of absence from work in the 12-month period following a health shock



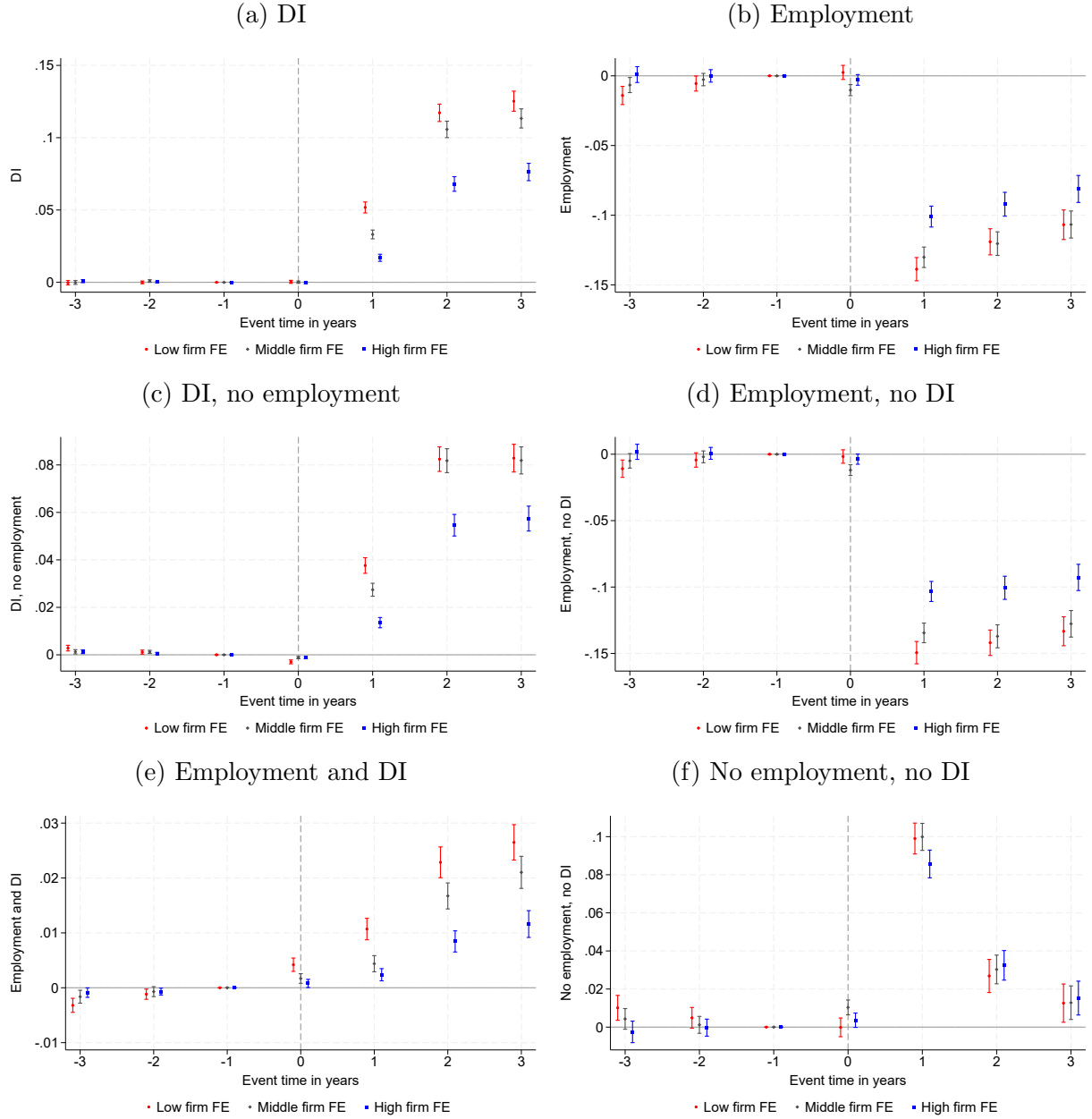
Note: Figure shows the distribution of hospital days (left panels) and days of absence (right panel) over the 12-month period following the health shock, by firm quality. Sample is restricted to people suffering a health shock, who were employed at the onset of the shock. The days of absence do not include the (unobserved) sick leave days, which are at most 15 days per calendar year. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A3: Health shock probability by firm quality



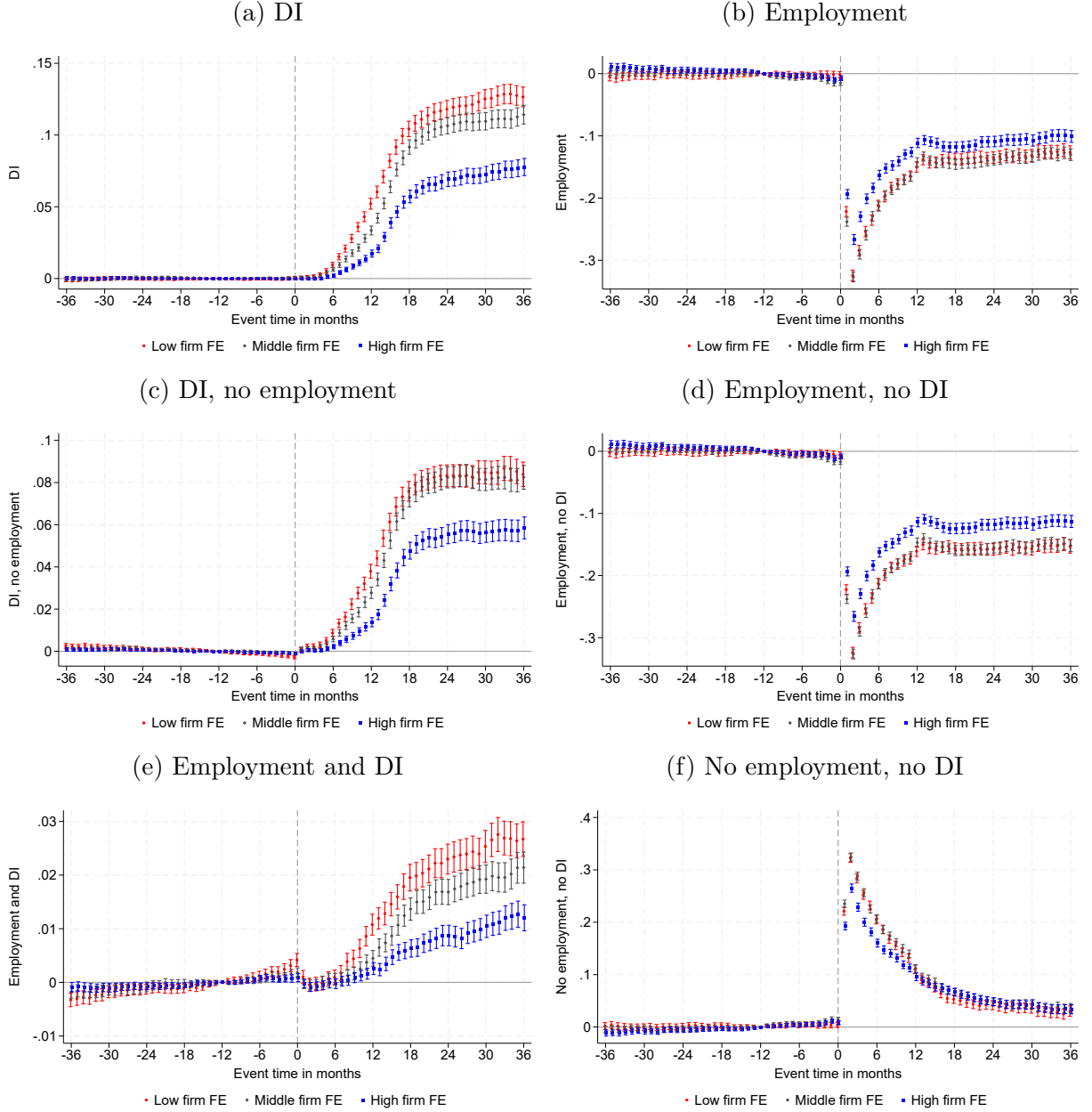
Note: Figure shows the probability of suffering a health shock by firm quality over 2011-2016. Panel (a) shows sample averages, Panel (b) shows regression estimates. Control variables in Panel (b) are: sex, age, age squared, 2-digit occupation code dummies, year dummies. 95% CI is displayed. Sample is restricted to employed people. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A4: Impact of health shocks over time



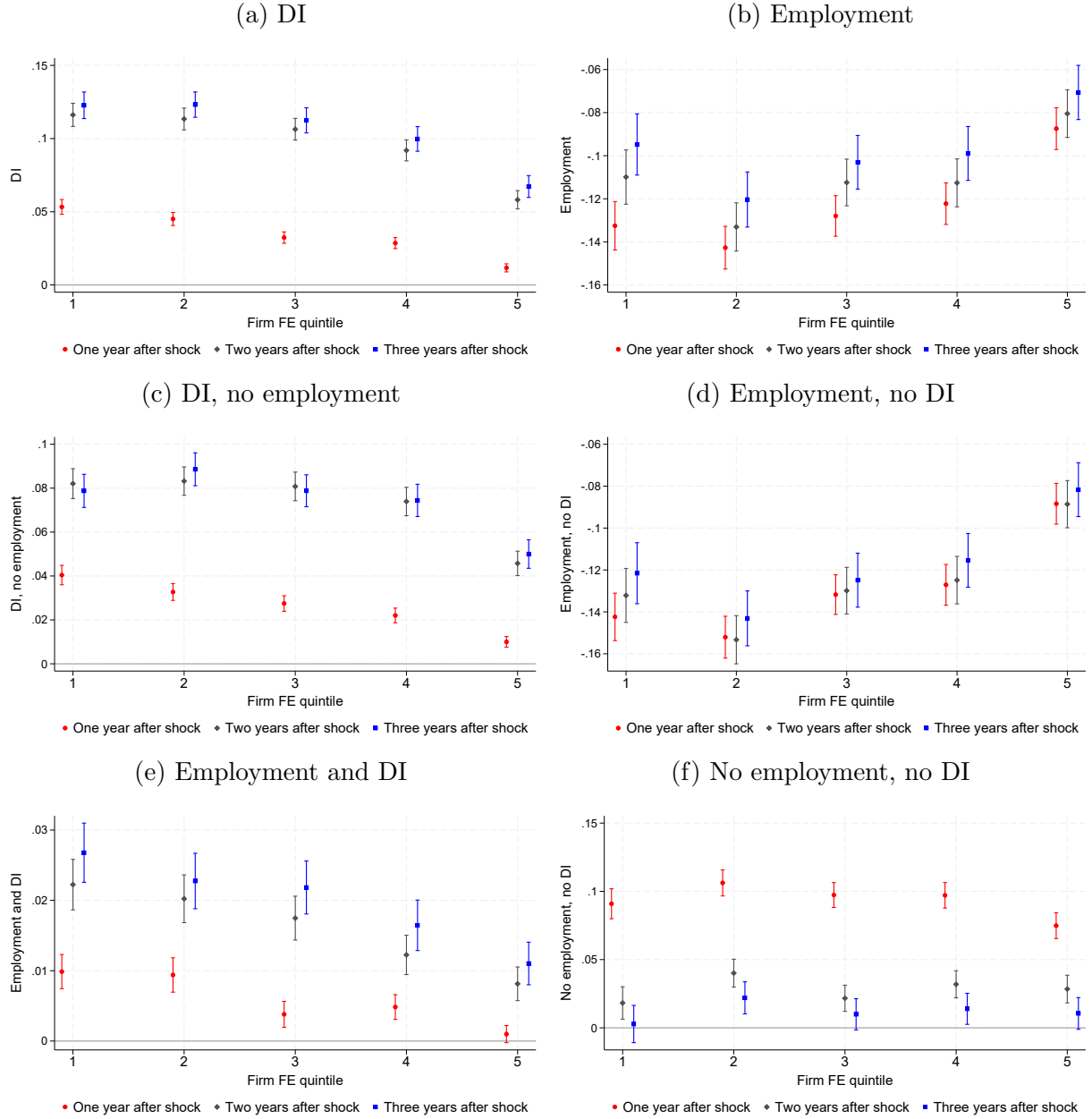
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A5: Impact of health shocks over time, monthly estimates



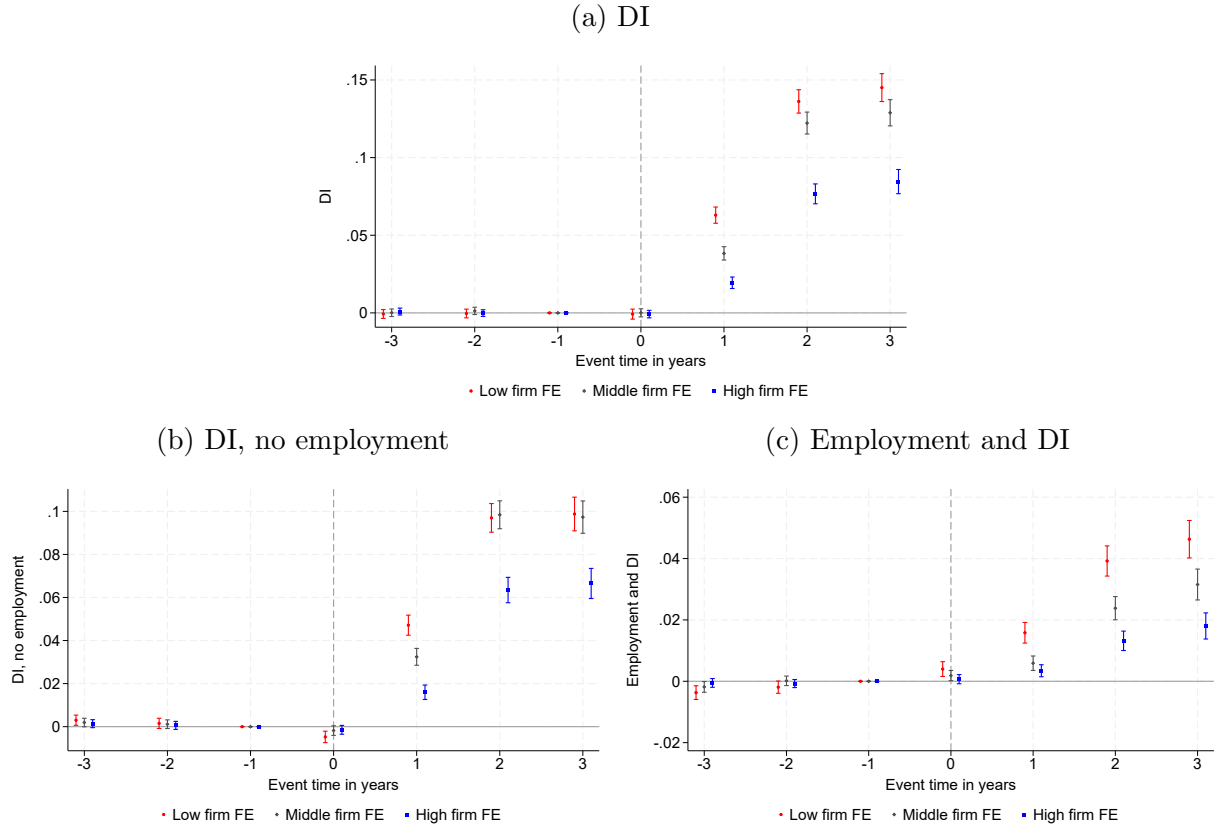
Note: Figure shows estimated β parameters of a modified Equation (3) with 95% confidence interval (based on robust standard errors). In this modified equation, we estimate the model on the monthly level but still use the 12th month before the shock as reference period. Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A6: Impact of health shocks by firm FE quintiles



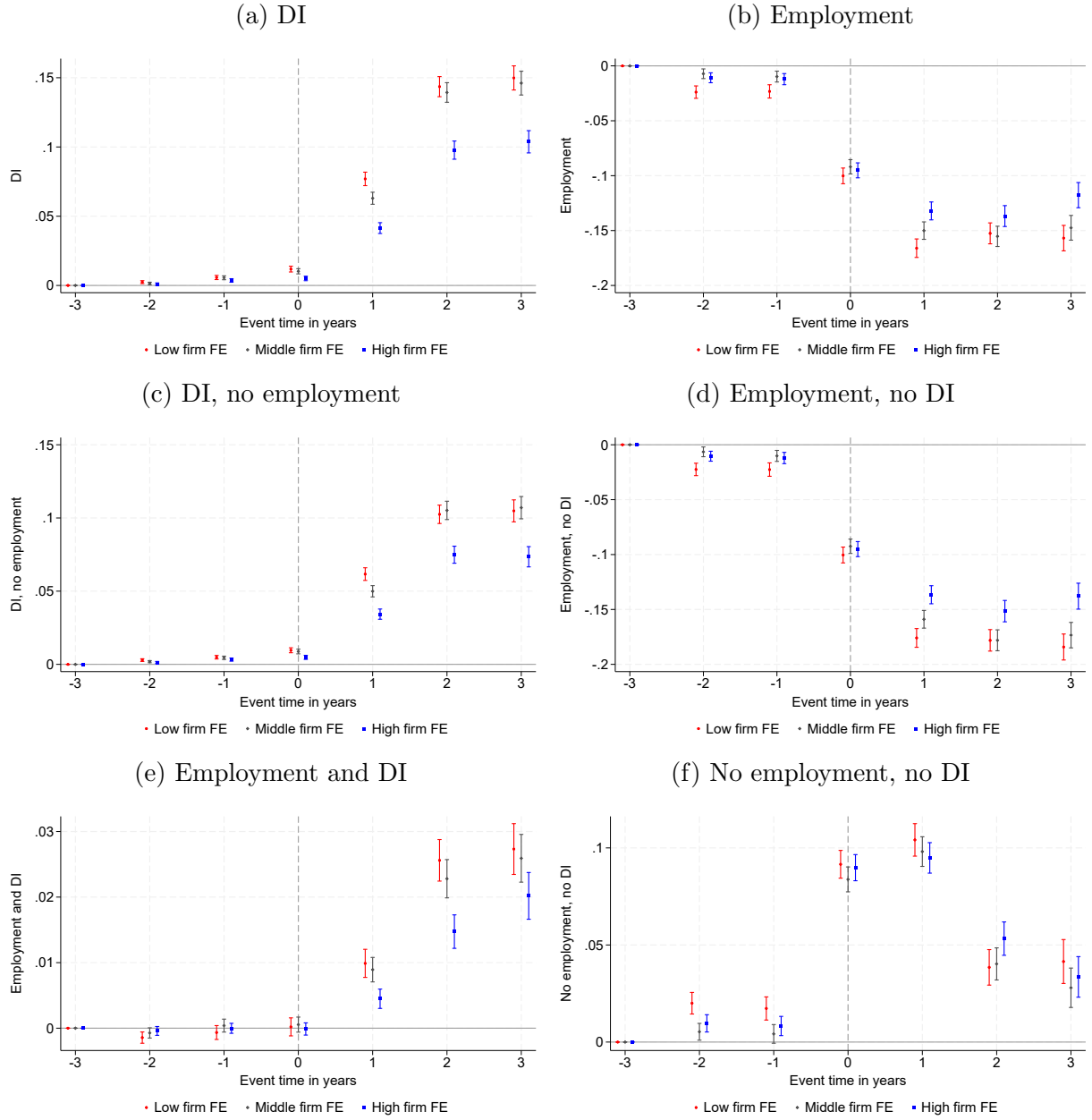
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to quintiles.

Appendix Figure A7: Impact of health shocks over time—estimation allowing for treatment effect heterogeneity



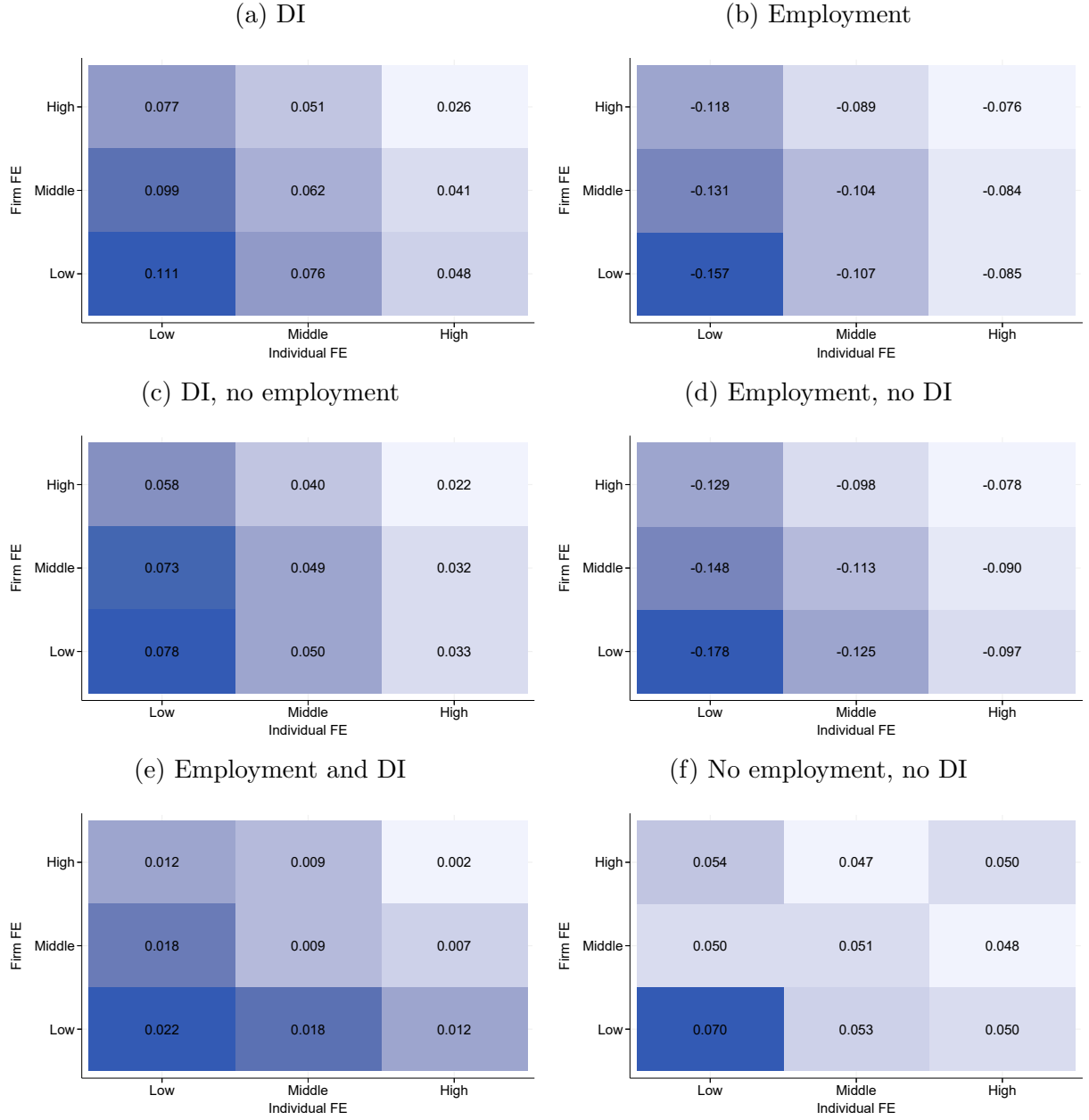
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). We use the estimator of Sun and Abraham (2021), implemented with Stata package Sun (2021). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A8: Impact of health shocks over time—firm quality measured at event time
–3



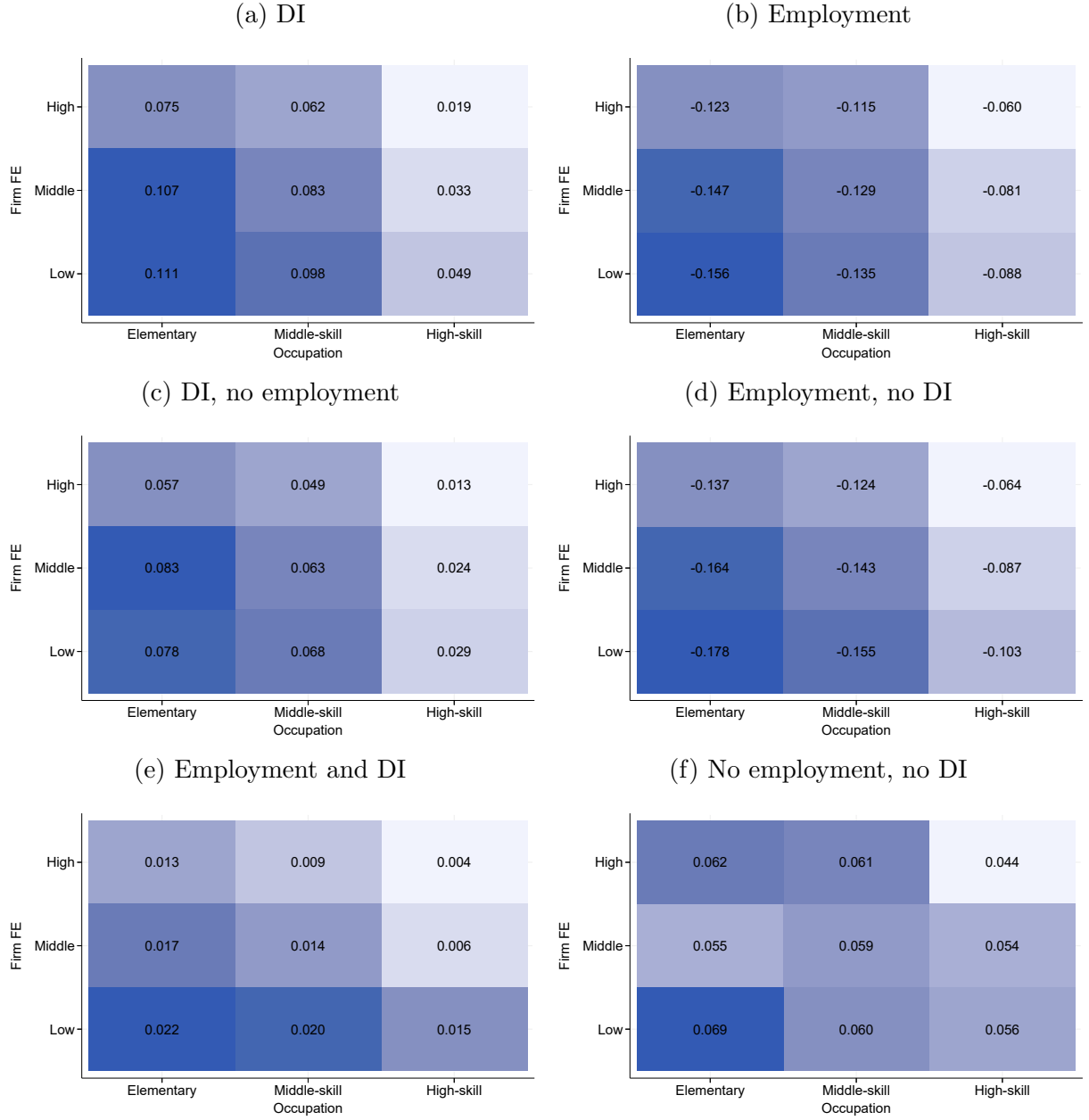
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time –3. Firm quality categories refer to the employer at event time –3. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A9: Average effect of health shocks by firm quality and individual quality



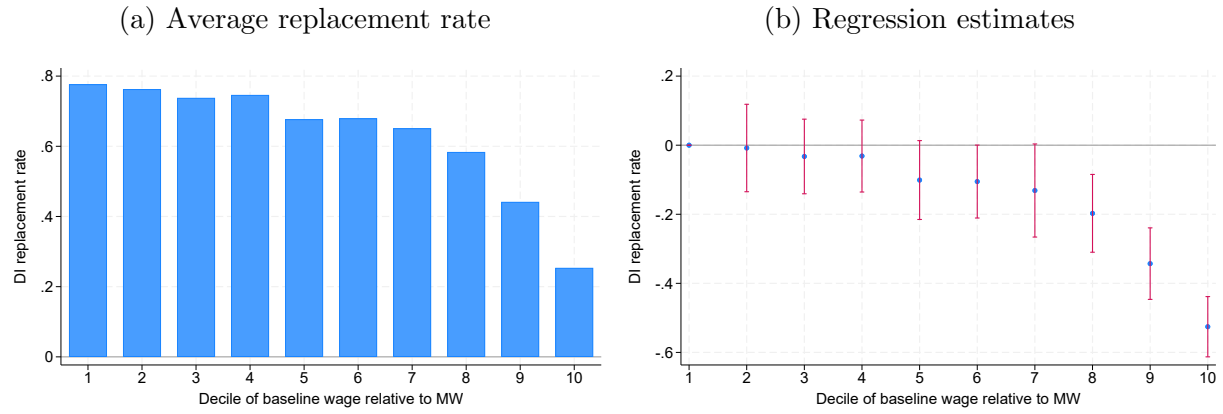
Note: Figure shows estimated β parameters of a modified version of Equation (2), in which the event time categories are replaced by a binary indicator that equals zero 1-3 years before the shock and equals one 1-3 years after the shock (the year of the event is omitted). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time 0. Individual quality categories refer to the individual at event time -1 . We measure individual and firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated individual and firm fixed effects to tertiles.

Appendix Figure A10: Average effect of health shocks by firm quality and occupation category



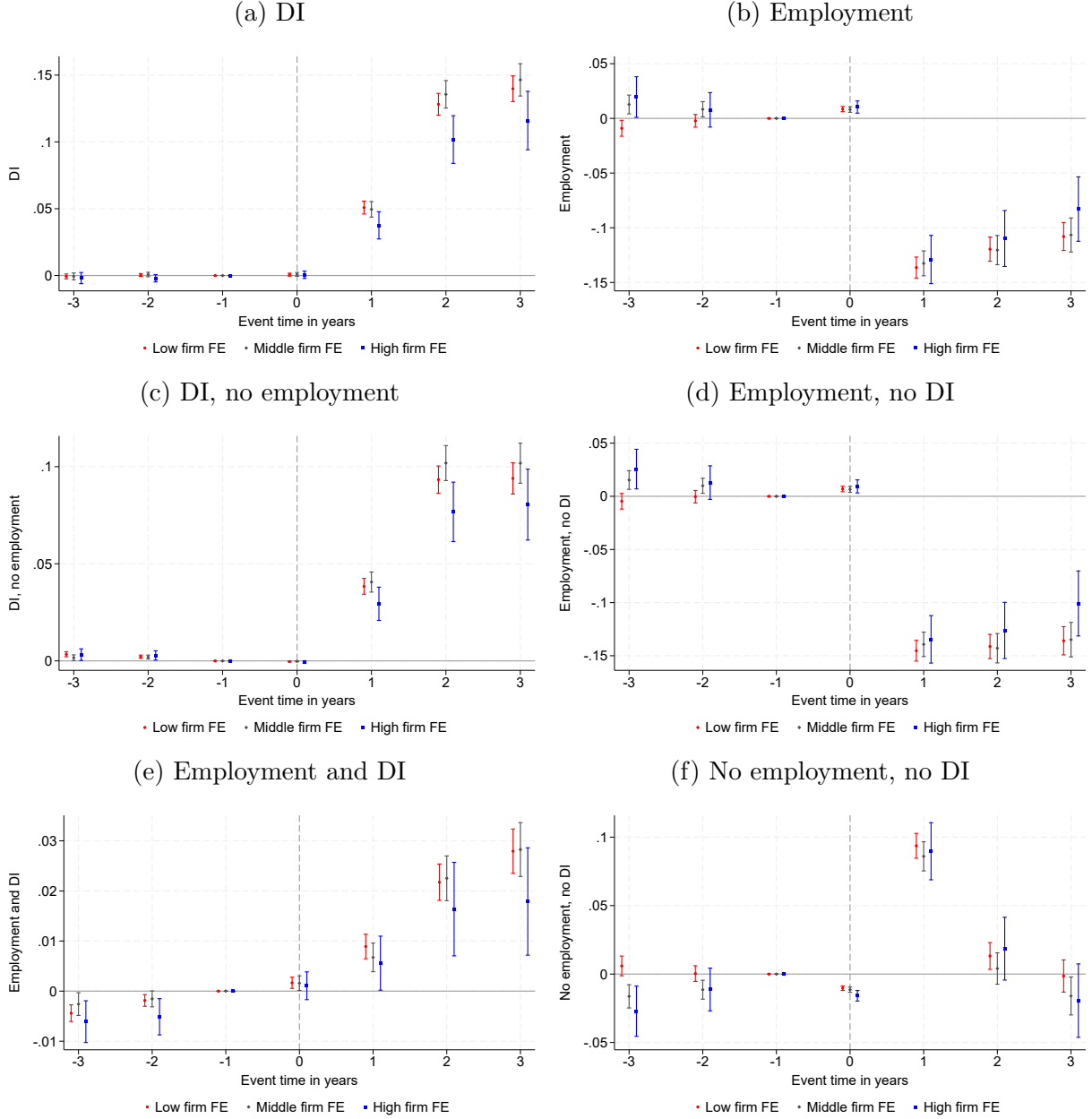
Note: Figure shows estimated β parameters of a modified version of Equation (2), in which the event time categories are replaced by a binary indicator that equals zero 1-3 years before the shock and equals one 1-3 years after the shock (the year of the event is omitted). Sample is restricted to people employed at event time zero. Firm quality and occupation categories refer to the employer at event time 0. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A11: DI wage replacement rate



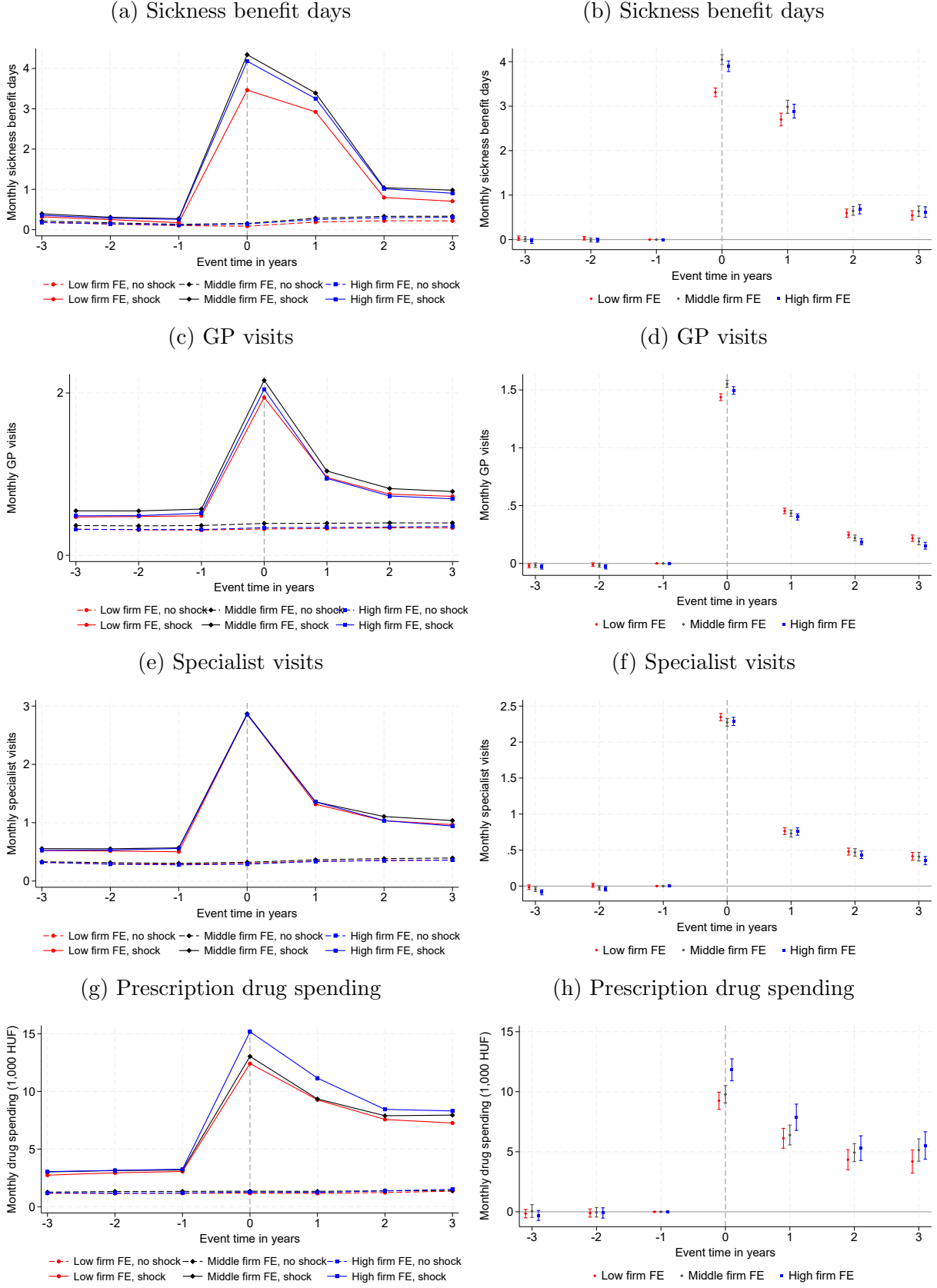
Note: Figure shows DI replacement rate by deciles of average of relative wage (relative to the minimum wage) 1-3 years before entering DI. To calculate the replacement rate, in the nominator we have the DI benefit measured at the first year of DI reciprocity, in the denominator we have the average wage income 1-3 years before entering DI. Panel (b) shows point estimates relative to the lowest decile with 95% confidence interval (based on robust standard errors). Control variables in Panel (b): calendar year, sex, age and age squared.

Appendix Figure A12: Impact of health shocks over time, low wage earners



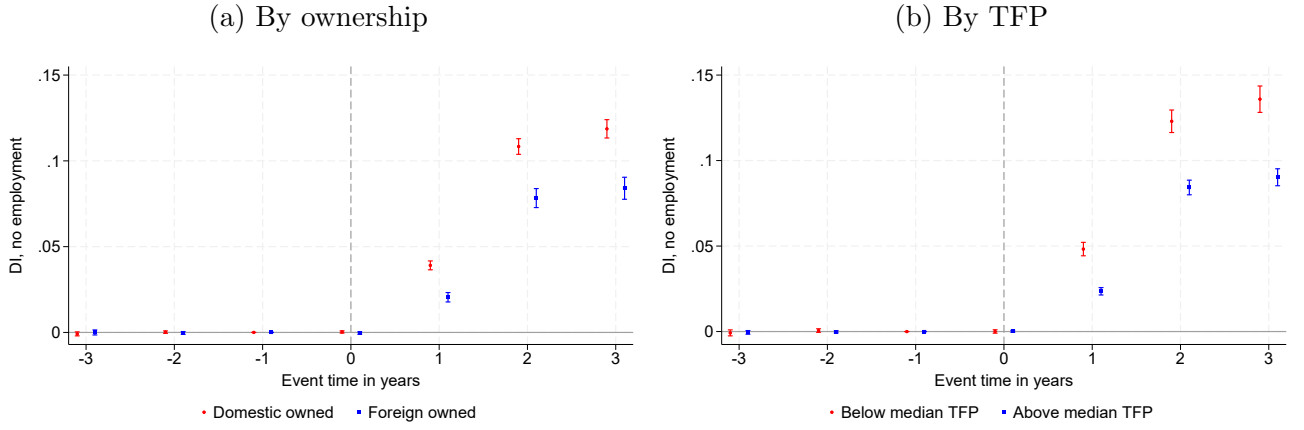
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero and 12 months before, whose wage relative to the minimum wage at event time -1 was in decile 1-5. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A13: Healthcare use over time



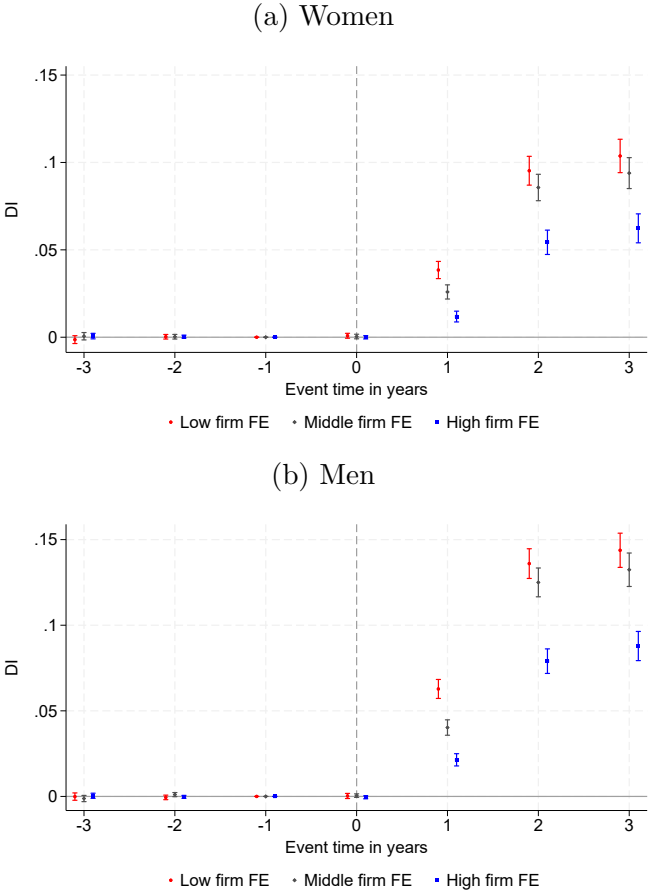
Note: Figure shows the time pattern of monthly healthcare use indicators. Left panels show means of healthcare use indicators, right panels show estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A14: Impact of health shocks on DI by ownership type and TFP



Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). The heterogeneity indicator is firm ownership (Panel (a)) and total factor productivity (TFP) (Panel (b)), both referring to the employer at event time zero. Sample is restricted to people employed at event time zero.

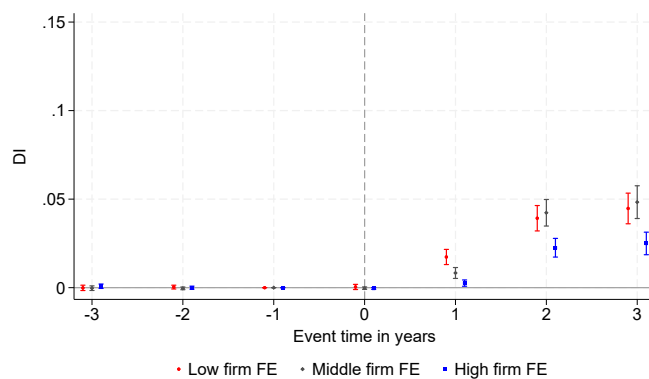
Appendix Figure A15: Impact of health shocks on DI over time by sex



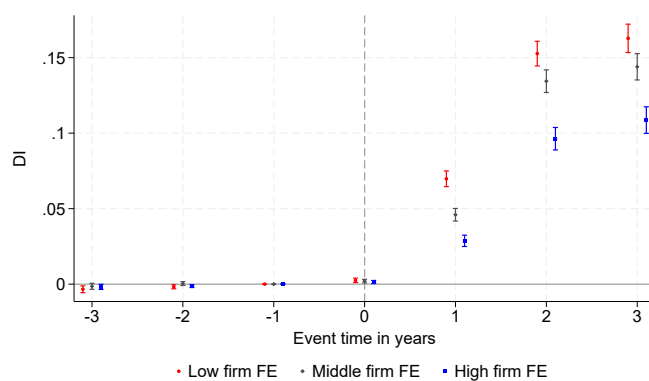
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A16: Impact of health shocks on DI over time by age of worker

(a) Age 18–39

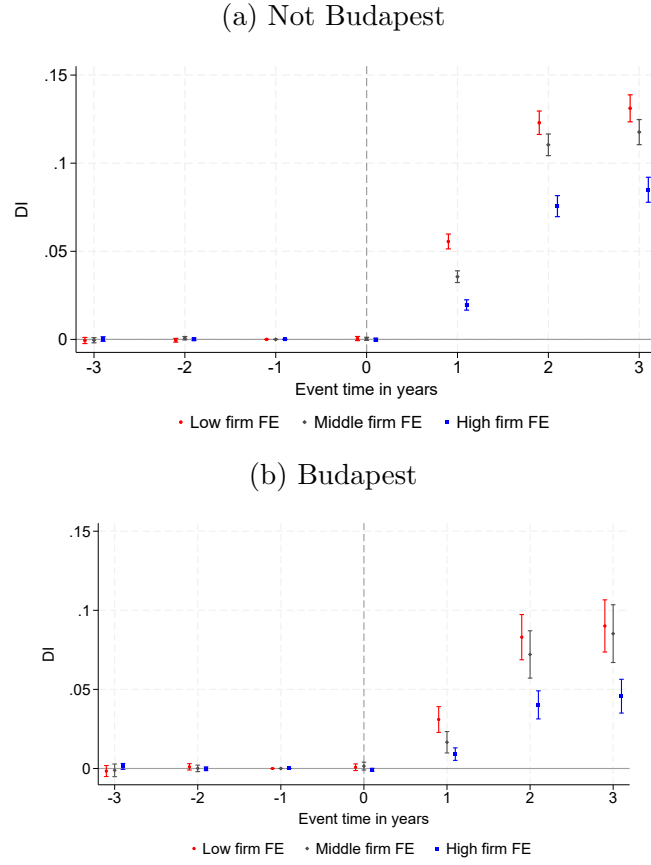


(b) Age 40–60



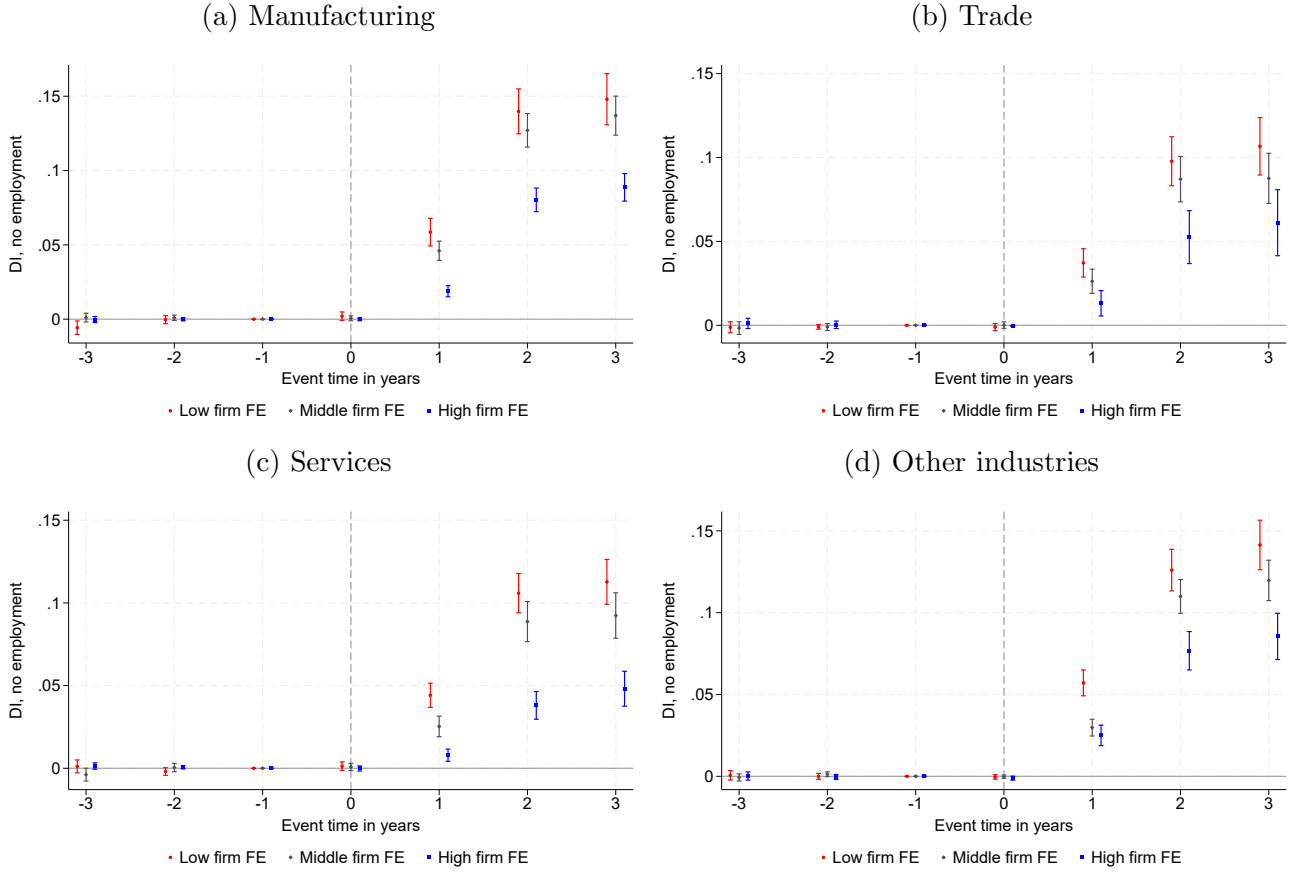
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is split by age (18 – 39 vs. 40 – 60). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A17: Impact of health shocks on DI over time by living area



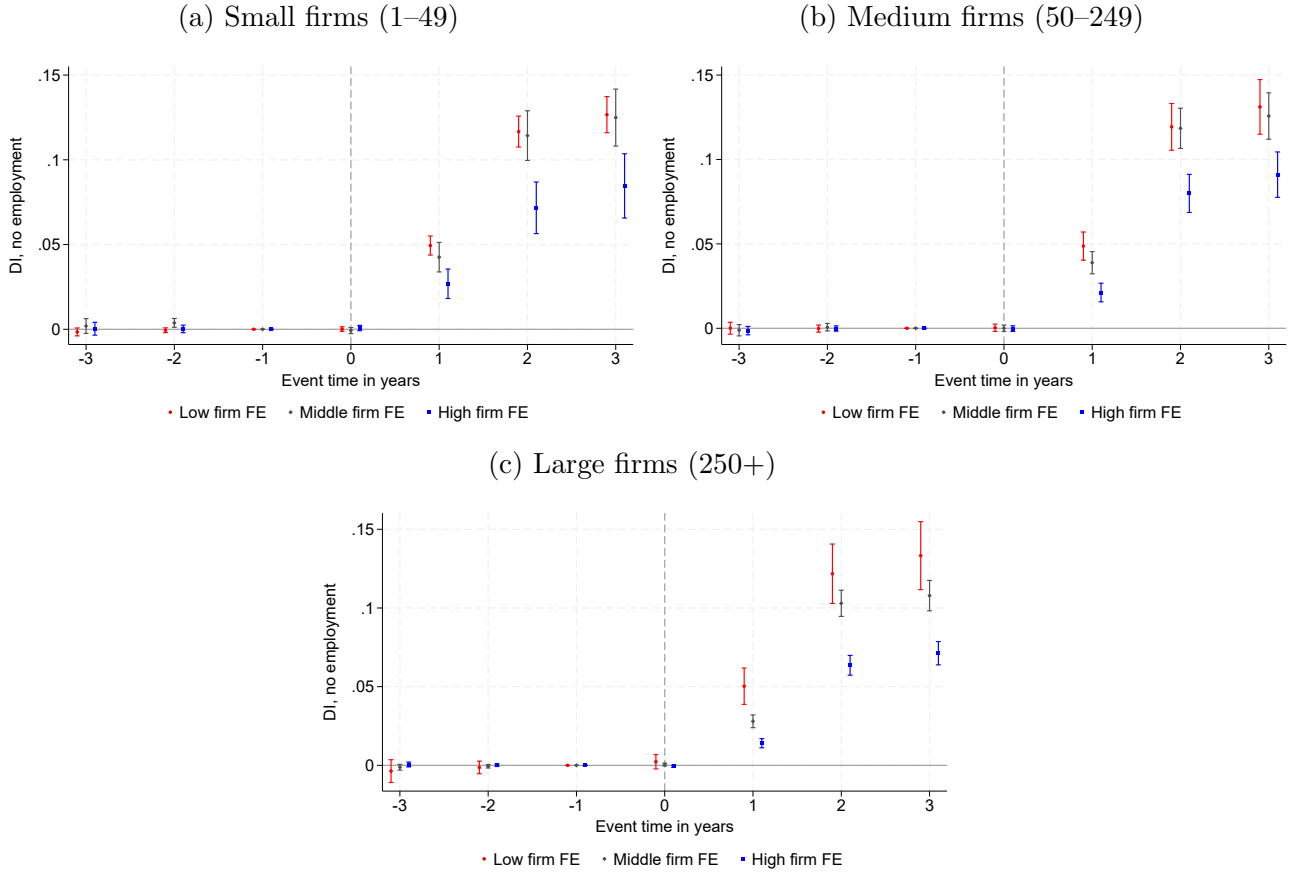
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is split by living area (Budapest or outside Budapest). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A18: Impact of health shocks on DI by industry groups and AKM firm FE categories



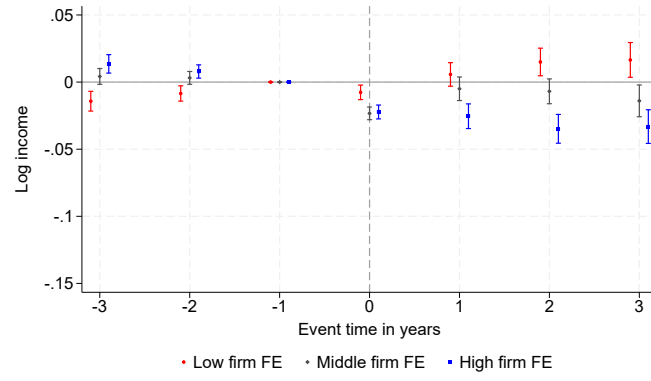
Note: Figure shows estimated parameters of the interaction term between the treatment indicator, event time, firm quality (AKM firm FE tertiles), and industry categories in an extended version of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Industry and firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A19: Impact of health shocks on DI by firm size and AKM firm FE categories



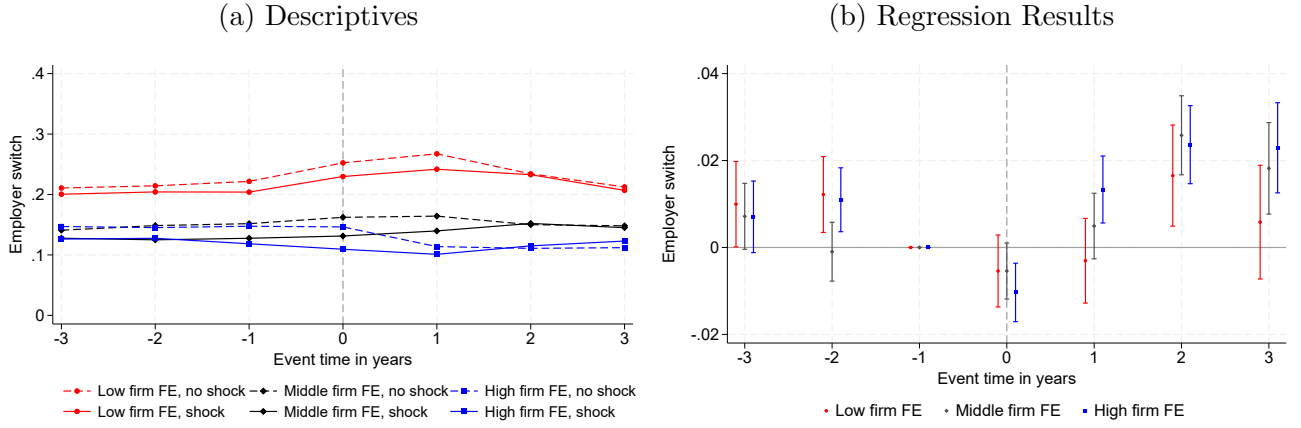
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval, splitting the sample by firm size category. Sample is restricted to people employed at event time zero. Firm size and quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A20: Impact of health shocks on income over time



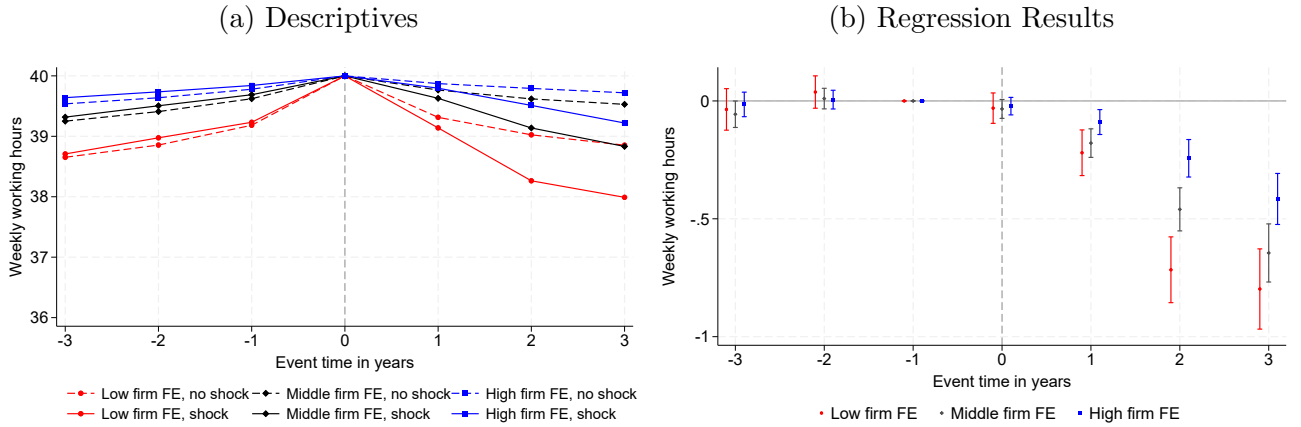
Note: Figure shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to people employed at event time zero. Firm quality categories refer to the employer at event time zero. Income is the sum of the annual average monthly wage and DI benefit. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A21: Impact of health shocks on switching employer, conditional on employment



Note: Figure shows the time pattern of the probability of moving between employers. The binary indicator of switching employer is defined on the sample of people employed currently and 12 months before. It equals one if the current employer and the employer 12 months before are different. Panel (a) shows averages of employer switches for individuals suffering a health shock (treated group) and individuals who never suffered a health shock (control group). Panel (b) shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to employed people. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Figure A22: Impact of health shocks on weekly working hours, conditional on employment



Note: Panel (a) shows averages of weekly working hours for individuals suffering a health shock (treated group) and individuals who never suffered a health shock (control group). Panel (b) shows estimated β parameters of Equation (3) with 95% confidence interval (based on robust standard errors). Sample is restricted to employed people. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Table A1: Prevalence of major disease groups at the onset of health shocks

	Firm FE tertile		
	Low	Middle	High
Accident	0.10	0.09	0.10
Cancer	0.11	0.11	0.11
Cardiovascular	0.19	0.18	0.17
Digestive	0.11	0.11	0.11
Mental	0.08	0.08	0.07
Musculoskeletal	0.13	0.14	0.14
Respiratory	0.06	0.06	0.06
Urogenital	0.09	0.10	0.12

Note: Table shows share of major categories of inpatient diagnosis codes (ICD-10 codes) in the first month of the health shock. The health shock definition is provided in Section 2. Sample is restricted to people employed at the onset of the shock. We split the table by firm FE tertiles – we measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles.

Appendix Table A2: Sorting of individuals across firm types

No health shock		Firm FE tertile			
Individual FE tertile		Low	Middle	High	Total
	Low	11%	13%	7%	31%
	Middle	12%	11%	10%	33%
	High	6%	10%	20%	36%
	Total	30%	33%	36%	100%
N=464,604					
With health shock		Firm FE tertile			
Individual FE tertile		Low	Middle	High	Total
	Low	14%	16%	9%	39%
	Middle	11%	11%	9%	30%
	High	6%	9%	15%	30%
	Total	31%	37%	32%	100%
N=47,806					

Note: Table shows the distribution of workers across individual and firm FE tertiles. Firm FE and individual FE are measured one year before the health shock or before the random event for the no health shock group. We measure individual and firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated individual and firm fixed effects to tertiles.

Appendix Table A3: Average effect of health shocks by disease categories

	DI	Employment	DI, no emp.	Emp. & DI	Emp., no DI	No emp., no DI
Accident						
Low firm FE	0.039*** [0.005]	-0.070*** [0.011]	0.023*** [0.004]	0.011*** [0.002]	-0.082*** [0.011]	0.041*** [0.011]
Middle firm FE	0.034*** [0.005]	-0.089*** [0.011]	0.030*** [0.004]	0.001 [0.002]	-0.090*** [0.011]	0.057*** [0.010]
High firm FE	0.019*** [0.004]	-0.072*** [0.010]	0.014*** [0.003]	0.004** [0.002]	-0.076*** [0.010]	0.056*** [0.009]
Cancer						
Low firm FE	0.216*** [0.011]	-0.229*** [0.013]	0.156*** [0.009]	0.042*** [0.005]	-0.271*** [0.014]	0.063*** [0.011]
Middle firm FE	0.170*** [0.009]	-0.240*** [0.012]	0.135*** [0.008]	0.027*** [0.004]	-0.267*** [0.012]	0.098*** [0.010]
High firm FE	0.111*** [0.009]	-0.166*** [0.013]	0.088*** [0.008]	0.016*** [0.003]	-0.181*** [0.013]	0.070*** [0.010]
Cardiovascular						
Low firm FE	0.160*** [0.007]	-0.155*** [0.009]	0.116*** [0.006]	0.029*** [0.003]	-0.184*** [0.009]	0.029*** [0.007]
Middle firm FE	0.124*** [0.006]	-0.139*** [0.008]	0.096*** [0.005]	0.020*** [0.002]	-0.159*** [0.008]	0.035*** [0.006]
High firm FE	0.090*** [0.006]	-0.107*** [0.009]	0.074*** [0.005]	0.011*** [0.002]	-0.118*** [0.009]	0.028*** [0.007]
Digestive						
Low firm FE	0.041*** [0.006]	-0.056*** [0.011]	0.024*** [0.004]	0.011*** [0.003]	-0.066*** [0.012]	0.024** [0.011]
Middle firm FE	0.030*** [0.005]	-0.061*** [0.010]	0.022*** [0.004]	0.006*** [0.002]	-0.066*** [0.011]	0.034*** [0.009]
High firm FE	0.022*** [0.004]	-0.029*** [0.010]	0.019*** [0.004]	0.001 [0.001]	-0.030*** [0.010]	0.009 [0.009]
Mental						
Low firm FE	0.057*** [0.007]	-0.143*** [0.013]	0.033*** [0.005]	0.015*** [0.003]	-0.158*** [0.013]	0.099*** [0.013]
Middle firm FE	0.053*** [0.006]	-0.146*** [0.012]	0.037*** [0.006]	0.012*** [0.003]	-0.159*** [0.013]	0.103*** [0.012]
High firm FE	0.034*** [0.006]	-0.137*** [0.014]	0.022*** [0.005]	0.008*** [0.003]	-0.145*** [0.014]	0.107*** [0.013]
Musculoskeletal						
Low firm FE	0.051*** [0.006]	-0.070*** [0.009]	0.026*** [0.004]	0.017*** [0.003]	-0.087*** [0.010]	0.040*** [0.009]
Middle firm FE	0.049*** [0.005]	-0.066*** [0.008]	0.027*** [0.004]	0.017*** [0.003]	-0.083*** [0.008]	0.035*** [0.007]
High firm FE	0.027*** [0.004]	-0.064*** [0.008]	0.020*** [0.003]	0.004** [0.002]	-0.068*** [0.009]	0.041*** [0.008]
Respiratory						
Low firm FE	0.071*** [0.009]	-0.085*** [0.015]	0.051*** [0.007]	0.014*** [0.004]	-0.099*** [0.016]	0.038*** [0.013]
Middle firm FE	0.065*** [0.008]	-0.095*** [0.014]	0.055*** [0.007]	0.007** [0.003]	-0.102*** [0.015]	0.032** [0.012]
High firm FE	0.035*** [0.007]	-0.047*** [0.013]	0.027*** [0.005]	0.003 [0.002]	-0.050*** [0.013]	0.013 [0.012]
Urogenital						
Low firm FE	0.040*** [0.005]	-0.138*** [0.012]	0.028*** [0.004]	0.007*** [0.002]	-0.144*** [0.012]	0.104*** [0.011]
Middle firm FE	0.036*** [0.004]	-0.132*** [0.010]	0.031*** [0.004]	0.004*** [0.001]	-0.135*** [0.011]	0.099*** [0.010]
High firm FE	0.015*** [0.003]	-0.121*** [0.009]	0.010*** [0.003]	0.003** [0.001]	-0.124*** [0.009]	0.107*** [0.009]

Note: Table shows estimated β parameters of a modified version of Equation (3), in which the event time categories are replaced by a binary indicator that equals zero 1-3 years before the shock and equals one 1-3 years after the shock (the year of the event is omitted). Sample is restricted to workers employed at event time zero. Firm quality categories refer to the employer at event time zero. We measure firm quality with an AKM-type fixed effect (Equation (1)) and divide the estimated firm fixed effects to tertiles. Robust standard errors in brackets. *** p<0.01, ** p<0.05, * p<0.1.